

A redshift-dependent External Field Effect as a decisive test of an evolving MOND scale: forecast and observing proposal

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Abstract

The Milgromian acceleration scale $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$ is numerically close to the cosmic acceleration cH_0 . If this is causal rather than coincidental— $a_0 = \frac{c}{2}\sqrt{G\rho} = cH(z)/Z$, so that the scale tracks the mean cosmic density—then a_0 *evolves*, $a_0(z) = a_0(0)E(z)$ with $E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$. We point out that this hypothesis makes one prediction that is both *forbidden in Λ CDM* and *distinct from constant- a_0 MOND*: the External Field Effect (EFE) must weaken with redshift, its strength scaling as $\eta(z) = g_{\text{ext}}/a_0(z) = \eta_0/E(z)$. We compute the observable signal (the change in the embedded-versus-isolated mass-discrepancy offset, ~ 0.1 dex out to $z \simeq 4$ –6), build an honest per-galaxy error budget ($\sigma \simeq 0.25$ –0.41 dex), and derive the required sample: ~ 590 (optimistic) to ~ 1600 (realistic) *extended, environment-resolved* galaxies with kinematics, baryonic masses and a measured external field, the discriminating power concentrated at $z \gtrsim 2$. We lay out a concrete, staged JWST + ALMA + ELT campaign. We are explicit that the present direct evidence for $a_0(z)$ is only a $\sim 2\sigma$, single-point-driven, Λ CDM-degenerate hint; the EFE-evolution test is the cleaner discriminator and a next-decade measurement.

1 Introduction

Modified Newtonian Dynamics (MOND; [1, 2]) organises galaxy dynamics around a single acceleration scale a_0 : where the gravitational acceleration falls below a_0 , the observed acceleration g_{obs} exceeds the Newtonian baryonic value g_{bar} , following the radial acceleration relation (RAR) with remarkably small scatter [3, 4]. The value of a_0 has a famous numerical property: $2\pi a_0 \approx cH_0 \approx c^2\sqrt{\Lambda/3}$. Two readings are possible— a_0 is a new fundamental constant, or it is set by cosmology [5, 6]. We take the second, in the specific form

$$a_0 = \frac{c}{2}\sqrt{G\rho} = \frac{cH(z)}{Z}, \quad Z = 2\sqrt{8\pi/3} \simeq 5.79, \quad (1)$$

where the second equality uses the Friedmann relation $H^2 = \frac{8\pi G}{3}\rho$. Reading ρ as the cosmic mean density makes a_0 a function of epoch:

$$a_0(z) = a_0(0) E(z), \quad E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \quad (2)$$

The evolving-scale idea is Milgrom’s [6]; what we contribute here is a single *clean, falsifiable* consequence and the survey design to test it. That consequence lives in the MOND External Field Effect.

The External Field Effect. MOND is nonlinear in acceleration, so—unlike Newtonian gravity or particle dark matter, which obey the strong equivalence principle—a *uniform external field* g_{ext} alters a system’s internal dynamics [7]. The controlling ratio is $\eta \equiv g_{\text{ext}}/a_0$: when $\eta \gtrsim 1$ the external field drives the system toward the Newtonian regime and the MOND boost is suppressed; when $\eta \ll 1$ the system is in isolated, deep-MOND. The EFE is arguably detected at $z \simeq 0$ in the outer rotation curves of SPARC galaxies [8], though the significance is debated. Our point is simple: in Eq. (2) the *denominator* of η grows with redshift, so

$$\boxed{\eta(z) = \frac{g_{\text{ext}}}{a_0(z)} = \frac{\eta_0}{E(z)}} \quad (3)$$

and the EFE must *weaken* toward high z : at fixed external field, distant galaxies are more isolated-MOND than their local analogues.

2 The prediction and what it separates

Figure 1 shows Eqs. (2)–(3). By $z = 4$, $E(z) = 6.3$, so a given external environment is $\sim 6\times$ less effective at suppressing MOND than today. Three hypotheses make qualitatively different statements about the embedded-minus-isolated offset in the mass discrepancy, $\Delta \equiv \log_{10}(\text{boost}_{\text{emb}}/\text{boost}_{\text{iso}})$, as a function of redshift:

- **Λ CDM / dark matter:** $\Delta(z) = 0$ at all z (no EFE; the equivalence principle holds for internal dynamics).
- **Constant- a_0 MOND:** $\Delta(z) = \text{const} \neq 0$ (EFE present, not evolving).
- **Evolving a_0 (this work):** $\Delta(z) \rightarrow 0$ as z grows, tracking $\eta_0/E(z)$.

This is the key virtue of the test: the *existence* of a nonzero Δ already discriminates MOND from dark matter, and its *redshift trend* discriminates evolving from constant a_0 . Because the EFE is forbidden in Λ CDM, the redshift trend cannot be mimicked by halo evolution—the principal degeneracy that afflicts a direct $a_0(z)$ measurement (§4). Figure 2 shows the three curves with forecast error bars.

3 The observable

For a galaxy of baryonic acceleration g_{bar} in an external field g_{ext} , the algebraic (single-field) MOND/AQUAL prescription with interpolation function $\mu(x) = x/(1+x)$ gives the internal acceleration g_{obs} by solving

$$\mu\left(\frac{g_{\text{obs}} + g_{\text{ext}}}{a_0(z)}\right) g_{\text{obs}} = g_{\text{bar}}. \quad (4)$$

The isolated limit ($g_{\text{ext}} = 0$) recovers $g_{\text{obs}} = \sqrt{g_{\text{bar}} a_0}$; a strong external field Newtonises the system, $g_{\text{obs}} \rightarrow g_{\text{bar}}/\mu(g_{\text{ext}}/a_0)$. The observable offset is $\Delta(z) = \log_{10}[(g_{\text{obs}}^{\text{emb}}/g_{\text{bar}})/(g_{\text{obs}}^{\text{iso}}/g_{\text{bar}})]$. For a deep-MOND galaxy ($g_{\text{bar}} = 0.1 a_0$) in a moderate field ($g_{\text{ext}} = 0.5 a_0$), Eq. (4) yields $\Delta = -0.196$ dex at $z = 0$, rising to -0.115 dex at $z = 4$; the full $z = 0 \rightarrow 6$ swing for $g_{\text{ext}} = a_0$ is ≈ 0.12 dex (Fig. 2). *This is the signal*—small, but with a fixed sign and a fixed redshift shape set entirely by $E(z)$.

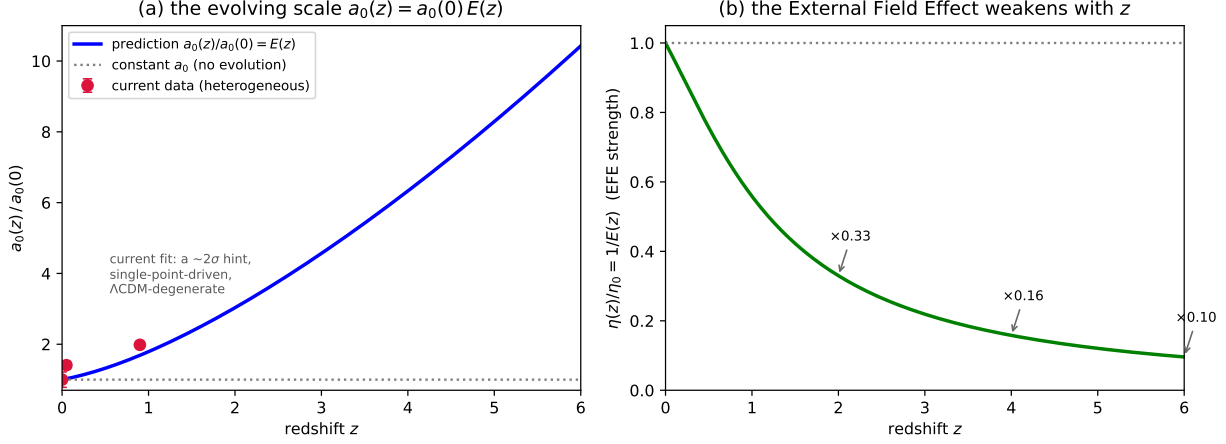


Figure 1: The prediction. (a) The evolving scale $a_0(z) = a_0(0)E(z)$ (blue), with the three heterogeneous determinations compiled to date (red; Table 1); the dotted line is a non-evolving a_0 . (b) The EFE strength $\eta(z)/\eta_0 = 1/E(z)$: the external field becomes $\sim 3\times$, $6\times$, $10\times$ less effective by $z = 2, 4, 6$.

4 Current evidence, stated honestly

Table 1 collects the three published acceleration-scale determinations spanning cosmic time. A power-law fit $a_0 \propto (1+z)^p$ gives $p = 0.80 \pm 0.17$, and a constant a_0 is nominally disfavoured at $\sim 5\sigma$. We do not believe that number. It rests on a single long-lever-arm point ($z \simeq 0.9$), it assumes three *heterogeneous* measurements (local disk rotation curves; an intermediate compilation; high- z integral-field dispersions) are commensurate to their small quoted errors, and folding in the inter-method systematic that the local pair already demands drops the rejection to $\sim 2\sigma$; removing the $z \simeq 0.9$ point drops it to $\sim 1.2\sigma$. Moreover a rising *apparent* a_0 is exactly what Λ CDM produces through denser high- z halos and more compact baryons [11, 12], so three points cannot separate fundamental $a_0(z)$ from astrophysics plus selection. **The honest status of the direct evidence is a $\sim 2\sigma$ hint in the predicted direction—not a detection.** This is precisely why the EFE-evolution test matters: it is the same physics seen through a window ($\Delta \neq 0$) that dark matter cannot enter.

z	a_0 [$10^{-10} \text{ m s}^{-2}$]	source
0.00	1.20 ± 0.26	SPARC—McGaugh, Lelli & Schombert 2016 [3]
0.05	1.69 ± 0.13	Văršeteanu <i>et al.</i> 2025 [16]
0.90	2.38 ± 0.11	MUSE–DARK III 2026 [17]

Table 1: The three acceleration-scale determinations used in Fig. 1. They are heterogeneous in method and sample; the apparent rise is a $\sim 2\sigma$ hint, Λ CDM-degenerate.

5 Forecast: signal, noise, sample size

Error budget. The per-galaxy uncertainty on $\log_{10}(M_{\text{dyn}}/M_{\text{bar}})$ at $z > 4$ is dominated by (i) dynamical mass from resolved kinematics (~ 0.20 dex), (ii) stellar mass from SED fitting (~ 0.25 dex; IMF, SFH, dust), (iii) gas mass, which can dominate the baryons and is rarely measured (~ 0.20 dex), and (iv) the irreducible RAR scatter (~ 0.15 dex). In quadrature, $\sigma_{\text{gal}} \simeq 0.41$ dex realistically, $\simeq 0.25$ dex in an optimistic floor with measured gas and tight SEDs.

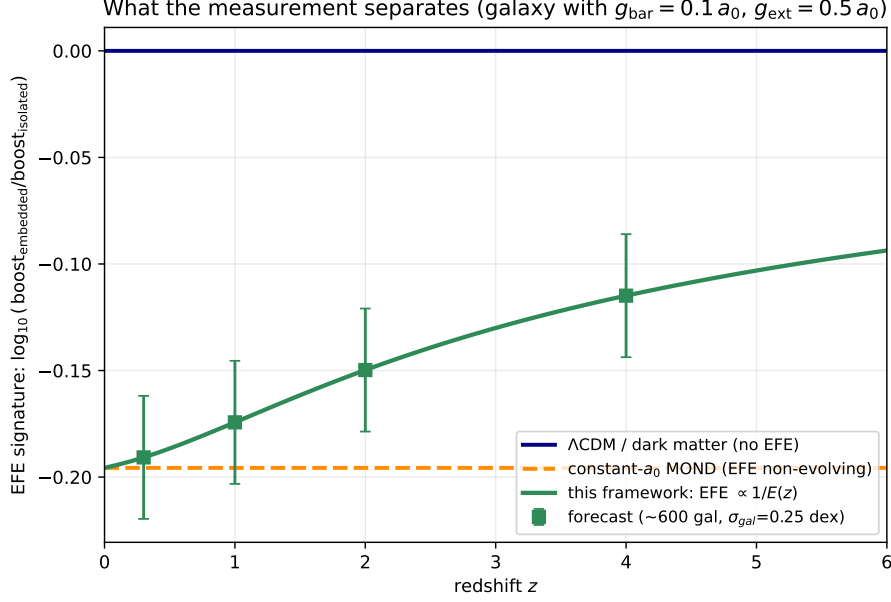


Figure 2: What the measurement separates. The embedded–isolated offset $\Delta(z)$ for a deep-MOND galaxy. Λ CDM predicts $\Delta = 0$ (no EFE); constant- a_0 MOND predicts a non-evolving offset (dashed); the evolving- a_0 hypothesis predicts $\Delta \rightarrow 0$ as $\eta_0/E(z)$ (green). Points: forecast for a ~ 600 -galaxy, $\sigma_{\text{gal}} = 0.25$ dex campaign.

Sample size. The distinctive test is the *double difference* $[\Delta_{\text{hi-}z} - \Delta_{\text{lo-}z}]$ across four sub-samples (isolated/embedded $\times$ low/high- z). With signal $\delta \simeq 0.12$ dex and $N/4$ galaxies per sub-sample, $\sigma(\text{double-diff}) = 2\sigma_{\text{gal}}/\sqrt{N/4}$, so a $k\sigma$ detection needs

$$N = \left(\frac{4k\sigma_{\text{gal}}}{\delta} \right)^2. \quad (5)$$

This gives $N \simeq 590$ (optimistic, 3σ) to $\simeq 1600$ (realistic, 3σ); a 5σ detection of the *evolution* needs ~ 1600 – 4400 (Fig. 3). The classic EFE *detection* (level i, $\Delta \neq 0$ at low z) is far cheaper—tens to a couple hundred galaxies—and is the near-term milestone; the *evolution* (level ii) is the hard, distinctive measurement.

6 The observing proposal

The campaign is a staged, multi-facility build of a *homogeneous* a_0 -versus-environment ladder from $z = 0$ to $z \gtrsim 4$. Every target must satisfy three selection criteria that are usually ignored: it must be **extended** (internal $g_{\text{bar}} < a_0(z)$, i.e. genuinely deep-MOND—compact galaxies are Newtonian and carry no signal); it must be **environment-classified** (isolated vs. embedded, with g_{ext} estimated from the local density field or host); and it must have a **full baryon census** (stars *and* gas).

Sequencing. (1) Re-analyse SPARC and existing VLT/JWST IFU surveys (KMOS^{3D}, KROSS, and JWST Cycle-1/2 kinematics) with a *single* pipeline and an environment split—this already tests level (i) and calibrates σ_{gal} . (2) A dedicated JWST+ALMA program for $1 < z < 4$ extended galaxies with gas masses. (3) ELT/HARMONI and ALMA deep fields for the $z > 4$ anchor. The decisive double-difference emerges only once the $z > 2.5$ bins are populated at the few-hundred level—an ELT-era, multi-cycle effort.

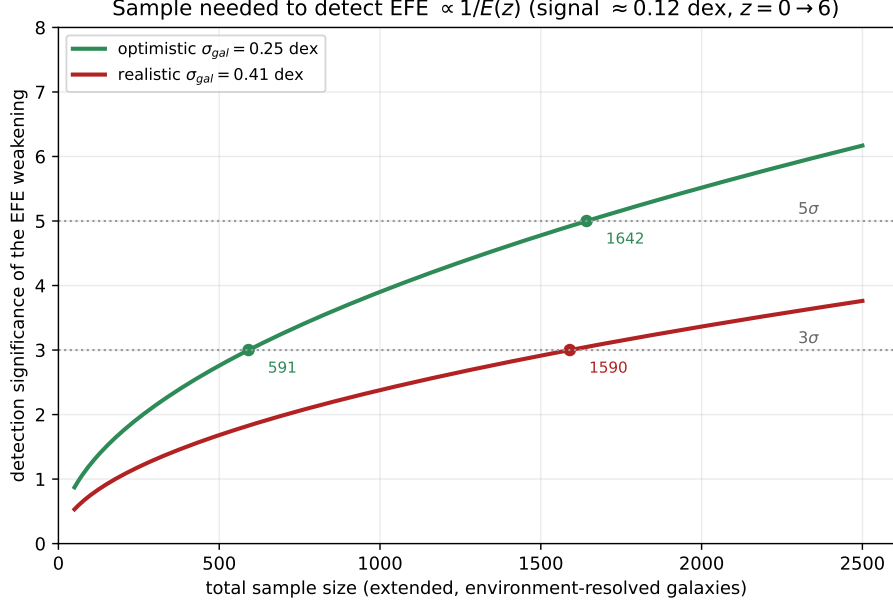


Figure 3: Sample size. Detection significance of the EFE *weakening* versus total number of extended, environment-resolved galaxies, for the realistic and optimistic per-galaxy noise. Markers give the 3σ and 5σ samples.

z bin	tracer	facility	N_{tgt}	role
0.0	H I /H α rotation	SPARC (archival)	~ 175	local anchor; RAR, EFE($z=0$)
0.3–0.8	H α IFU	VLT/MUSE, KMOS	~ 150	establish EFE; environment split
1–2.5	H α /Pa α IFU	JWST/NIRSpec	~ 150	a_0 , $M_\star$, environment
2.5–4	[CII]/CO disks	ALMA	~ 150	rotation, M_{gas} , M_{dyn}
> 4	[CII] + rest-optical	ALMA + ELT/HARMONI	~ 150	extended deep-MOND; the lever arm

Table 2: A staged EFE-versus- z campaign. The low- z bins establish that the EFE exists and pin systematics; the $z \gtrsim 2.5$ bins, where $E(z) \gtrsim 3$, carry the discriminating power. Total ~ 750 (optimistic floor) scaling to ~ 1600 if the realistic noise holds.

7 Caveats and degeneracies

We state the risks plainly.

1. **The premise chain is long.** The test is conditional on (a) MOND being a real description—contested by wide-binary analyses, which split between a Newtonian preference [13] and a MOND-consistent signal [14]; (b) the a_0 -scale being universal enough to define, itself debated [15]; and (c) the EFE being real at $z = 0$ [8]. Each link is uncertain.
2. **Environments evolve.** g_{ext} is not fixed in physical units across cosmic time; the analysis must *measure* g_{ext} per galaxy (local density, host mass) and match it across z , not assume it. This is a systematic, not a statistical, limitation.
3. **Selection.** High- z kinematic samples favour compact, dispersion-supported systems that are Newtonian and signal-free; the campaign must deliberately target extended, rotation-supported, low-acceleration galaxies.
4. **Baryon census.** Gas can dominate the high- z baryon budget; without M_{gas} , g_{bar} and hence

the EFE offset are biased.

None of these is fatal, but together they make this a careful, next-generation measurement rather than a re-analysis of existing data.

8 Conclusion

If the Milgromian scale is cosmological, $a_0 = \frac{c}{2}\sqrt{G\rho} = cH(z)/Z$, then the External Field Effect must fade with redshift as $\eta_0/E(z)$. This is the one prediction of the evolving- a_0 hypothesis that is simultaneously *forbidden in dark matter* and *distinct from constant- a_0 MOND*, and it is therefore the cleanest available discriminator—cleaner than the direct $a_0(z)$ measurement, whose present $\sim 2\sigma$ hint is Λ CDM-degenerate. The signal is small (~ 0.1 dex) and the required sample large (~ 590 – 1600 extended, environment-resolved galaxies, weighted to $z \gtrsim 2$), placing a decisive test in the JWST + ALMA + ELT next decade. A robust detection would establish two things at once: that the equivalence principle is violated in galaxy dynamics as MOND predicts, and that its acceleration scale is set by the expanding universe—evidence for an emergent, cosmologically-rooted origin of inertia. A null result for the *evolution*, given a low- z EFE, would falsify the cosmological reading and return a_0 to the status of a constant. Either way the question is decidable, and the experiment is specified.

Reproducibility. The signal, error budget, sample sizes and all three figures are generated by `reviews/efe_evolution_forecast.py` and `papers/efe_forecast_figures.py`; the current-data fit by `reviews/stresstest_piece3_evolution.py`.

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